

DAY-7

```
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JupyterLab Python [conda envbase] * Anaconda Toolbox

[8]: import pandas as pd
import seaborn as sns
#load titanic dataset
df=sns.load_dataset("titanic")
print(df.head())
print(df.shape)
print(df.info)

survived  pclass  sex  age  sibsp  parch  fare  embarked  class \
0         0       3  male  22.0    1     0   7.2500    S  Third
1         1       1  female  38.0    1     0  71.2833    C  First
2         1       3  female  26.0    0     0   7.9250    S  Third
3         1       1  female  35.0    1     0  53.1000    S  First
4         0       3  male   35.0    0     0   8.0500    S  Third

who  adult_male  deck  embark_town  alive  alone
0  man         True  NaN  Southampton  no  False
1  woman       False   C  Cherbourg  yes  False
2  woman       False  NaN  Southampton  yes  True
3  woman       False   C  Southampton  yes  False
4  man         True  NaN  Southampton  no  True
(891, 15)
<bound method DataFrame.info of
survived  pclass  sex  age  sibsp  parch  fare  embarked  class \
0         0       3  male  22.0    1     0   7.2500    S  Third
1         1       1  female  38.0    1     0  71.2833    C  First
2         1       3  female  26.0    0     0   7.9250    S  Third
3         1       1  female  35.0    1     0  53.1000    S  First
4         0       3  male   35.0    0     0   8.0500    S  Third
..         ...     ...     ...     ...     ...     ...     ...
886        0       2  male  27.0    0     0  13.0000    S  Second
887        1       1  female  19.0    0     0  30.0000    S  First
```

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..         ...     ...     ...     ...     ...     ...     ...
886        0       2  male  27.0    0     0  13.0000    S  Second
887        1       1  female  19.0    0     0  30.0000    S  First
888        0       3  female  NaN    1     2  23.4500    S  Third
889        1       1  male   26.0    0     0  30.0000    C  First
890        0       3  male   32.0    0     0   7.7500    Q  Third

who  adult_male  deck  embark_town  alive  alone
0  man         True  NaN  Southampton  no  False
1  woman       False   C  Cherbourg  yes  False
2  woman       False  NaN  Southampton  yes  True
3  woman       False   C  Southampton  yes  False
4  man         True  NaN  Southampton  no  True
..         ...     ...     ...     ...     ...
886  man         True  NaN  Southampton  no  True
887  woman       False   B  Southampton  yes  True
888  woman       False  NaN  Southampton  no  False
889  man         True   C  Cherbourg  yes  True
890  man         True  NaN  Queenstown  no  True

[891 rows x 15 columns]>

[9]: # THIS STEP IS CREATING ERROR IN THE DATASET LIKE IN EMBARK_TOWN CELL WE ARE CREATING 3 DIFF TYPES OF FORMATING FOR THE SAME WORD SO IT MAKES AN ERROR
df.loc[0,'embark_town']='southampton'
df.loc[1,'embark_town']='SOUTHAMPTON'
df.loc[2,'embark_town']='Southampton'
print(df['embark_town'].unique())

['southampton' 'SOUTHAMPTON' 'Southampton' 'Queenstown' 'Cherbourg' nan]

[11]: # TO CORRECT THE ERROR
# IF WE USE TITLE() IT CAPITALIZE THE 1ST LETTER SO ALL BECOMES SAME
df['embark_town']=df['embark_town'].str.title()
```

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[12]: #unique() prints the unique values and we found that there is a nan val
print(df['embark_town'].unique())

['Southampton' 'Queenstown' 'Cherbourg' nan]

[13]: #so we are counting the no of nan values
print(df['embark_town'].isnull().sum())

2

[14]: #using fillna fillling the nan value
df['embark_town']=df['embark_town'].fillna('Kingstown')

[15]: print(df['embark_town'].unique())

['Southampton' 'Queenstown' 'Cherbourg' 'Kingstown']

[17]: duplicate_row=df.iloc[5]
df_dup=pd.concat([df,duplicate_row])
print(df_dup.shape)

(896, 15)

[18]: print(df_dup.duplicated().sum())

112

[19]: #view duplicated rows
df_dup[df_dup.duplicated()]

[19]: survived  pclass  sex  age  sibsp  parch  fare  embarked  class  who  adult_male  deck  embark_town  alive  alone
47         1       3  female  NaN    0     0   7.7500    Q  Third  woman  False  NaN  Queenstown  yes  True
```

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```
[19]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
47	1	3	female	NaN	0	0	7.7500	Q	Third	woman	False	NaN	Queenstown	yes	True
76	0	3	male	NaN	0	0	7.8958	S	Third	man	True	NaN	Southampton	no	True
77	0	3	male	NaN	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True
87	0	3	male	NaN	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True
95	0	3	male	NaN	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Southampton	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

112 rows x 15 columns

```
[21]:
```

```
#drop duplicates/remove
df_dupsdf_dup.drop_duplicates()
print(df_dupsdf_dup.duplicated().sum())
0
```

```
[22]:
```

```
#CREATING SPELLING MISTAKES Bcz IN THE DS WE DONT HAVE SPELLING MISTAKE
df.loc[0,'embark_town']='Southampton'
df.loc[1,'embark_town']='Southampton'
print(df['embark_town'].value_counts())
```

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```
print(df['embark_town'].value_counts())
```

```
embark_town
Southampton    643
Cherbourg       167
Queenstown      77
Kingstown        2
Southampton      1
Southampton      1
Name: count, dtype: int64
```

```
[24]:
```

```
# fix typos
df['embark_town']=df['embark_town'].replace({
    'Southanton':'Southampton',
    'Southampton':'Southampton'
})
#verify:
print(df['embark_town'].value_counts())
```

```
embark_town
Southampton    645
Cherbourg       167
Queenstown      77
Kingstown        2
Name: count, dtype: int64
```

```
[27]:
```

```
#detect possible types
#drops null val and print the unique result
print(sorted(df['embark_town'].dropna().unique()))
['Cherbourg', 'Kingstown', 'Queenstown', 'Southampton']
```

```
[28]:
```

```
#creating problem here for suppose ages are accidentally entered in months insted of year
df.loc[0,'age']=240
df.loc[1,'age']=300
```

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```
df.loc[0,'age']=240
df.loc[1,'age']=300
```

```
[29]:
```

```
#detect unusual values
df['age'].describe()
```

```
[29]:
```

```
count    714.000000
mean      30.371387
std       19.362012
min        0.420000
25%       20.125000
50%       28.000000
75%       38.750000
max       300.000000
Name: age, dtype: float64
```

```
[30]:
```

```
#fix unit mismatch
#convert months to years
df.loc[df['age']>120,'age']=(
    df.loc[df['age']>120,'age']/12
)
#verify
df['age'].describe()
```

```
[30]:
```

```
count    714.000000
mean     29.678109
std      14.525902
min       0.420000
25%       20.000000
50%       28.000000
75%       38.000000
max       80.000000
Name: age, dtype: float64
```

```
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[32]: df['survived_label'] = df['survived'].map({
      0: 'No',
      1: 'Yes'
    })
      print(df.head())

      survived  pclass  sex  age  sibsp  parch  fare  embarked  class  \
0            0      3  male  20.0    1    0    7.2500    S  Third
1            1      1 female  25.0    1    0   71.2833    C  First
2            1      3 female  26.0    0    0    7.9250    S  Third
3            1      1 female  35.0    1    0   53.1000    S  First
4            0      3  male  35.0    0    0    8.0500    S  Third

      who  adult_male  deck  embark_town  alive  alone  survived_label
0    man         True   NaN  Southampton    no   False         No
1  woman        False    C  Southampton    yes   False         Yes
2  woman        False   NaN  Southampton    yes    True         Yes
3  woman        False    C  Southampton    yes   False         Yes
4    man         True   NaN  Southampton    no    True         No

[33]: #check: the df contains null values getting it in average so we use in mean
      df.isnull().mean()

[33]: survived      0.000000
      pclass      0.000000
      sex        0.000000
      age      0.198653
      sibsp      0.000000
      parch      0.000000
      fare      0.000000
      embarked  0.002245
      class      0.000000
      who        0.000000
      adult_male 0.000000
      dtype: float64
```

```
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      adult_male  0.000000
      deck      0.772166
      embark_town 0.000000
      alive      0.000000
      alone      0.000000
      survived_label 0.000000
      dtype: float64

[36]: #deck has more num of null val so we are deleting it bcz with null val we cant do anything in predicting or analysing
      #remove
      df.drop(columns=['deck'], inplace=True)

[37]: df.isnull().sum()

[37]: survived      0
      pclass      0
      sex        0
      age        0
      sibsp      0
      parch      0
      fare        0
      embarked    2
      class      0
      who        0
      adult_male  0
      embark_town 0
      alive      0
      alone      0
      survived_label 0
      dtype: int64

[40]: #replace missing values
      df['age'] = df['age'].fillna(df['age'].mean())
      df.head(20)
```

```
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[40]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	embark_town	alive	alone	survived_label
0	0	3	male	20.000000	1	0	7.2500	S	Third	man	True	Southampton	no	False	No
1	1	1	female	25.000000	1	0	71.2833	C	First	woman	False	Southampton	yes	False	Yes
2	1	3	female	26.000000	0	0	7.9250	S	Third	woman	False	Southampton	yes	True	Yes
3	1	1	female	35.000000	1	0	53.1000	S	First	woman	False	Southampton	yes	False	Yes
4	0	3	male	35.000000	0	0	8.0500	S	Third	man	True	Southampton	no	True	No
5	0	3	male	29.678109	0	0	8.4583	Q	Third	man	True	Queenstown	no	True	No
6	0	1	male	54.000000	0	0	51.8625	S	First	man	True	Southampton	no	True	No
7	0	3	male	2.000000	3	1	21.0750	S	Third	child	False	Southampton	no	False	No
8	1	3	female	27.000000	0	2	11.1333	S	Third	woman	False	Southampton	yes	False	Yes
9	1	2	female	14.000000	1	0	30.0708	C	Second	child	False	Cherbourg	yes	False	Yes
10	1	3	female	4.000000	1	1	16.7000	S	Third	child	False	Southampton	yes	False	Yes
11	1	1	female	58.000000	0	0	26.5500	S	First	woman	False	Southampton	yes	True	Yes
12	0	3	male	20.000000	0	0	8.0500	S	Third	man	True	Southampton	no	True	No
13	0	3	male	39.000000	1	5	31.2750	S	Third	man	True	Southampton	no	False	No
14	0	3	female	14.000000	0	0	7.8542	S	Third	child	False	Southampton	no	True	No
15	1	2	female	55.000000	0	0	16.0000	S	Second	woman	False	Southampton	yes	True	Yes
16	0	3	male	2.000000	4	1	29.1250	Q	Third	child	False	Queenstown	no	False	No
17	1	2	male	29.678109	0	0	13.0000	S	Second	man	True	Southampton	yes	True	Yes

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17	1	2	male	29.678109	0	0	13.0000	S	Second	man	True	Southampton	yes	True	Yes
18	0	3	female	31.000000	1	0	18.0000	S	Third	woman	False	Southampton	no	False	No
19	1	3	female	29.678109	0	0	7.2250	C	Third	woman	False	Cherbourg	yes	True	Yes

```
[49]: #Method 1: Mean Imputation
#Replace missing values with the average values
df['age']=df['age'].fillna(df['age'].median())
df.head(20)
```

```
[49]: age fare
0 22.0 7.25
1 25.0 8.05
2 23.5 7.90
3 24.0 8.00
4 23.0 7.50
```

```
[ ]: #method 3 :MODE
#Replace missing values with most frequent values
df['embarked_mode']=df['embarked'].fillna(df['embarked'].mode()[0])
```

```
[42]: #Method 4
#FORWARD FILL
print(df['embarked'].isnull().sum())
2
```

1.6.1.1. #apply forward fill

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```
[44]: #apply forward fill
df['embarked']=df['embarked'].ffill()
#check missing val
print(df['embarked'].isnull().sum())
0
```

```
[45]: import pandas as pd
import numpy as np
data={
    'age':[22,25,np.nan,24,23],
    'fare':[7.25,8.05,7.90,8.00,7.50]
}
df=pd.DataFrame(data)
print(df)
```

```
age fare
0 22.0 7.25
1 25.0 8.05
2 NaN 7.90
3 24.0 8.00
4 23.0 7.50
```

```
[48]: #method 5
#apply knn imputation
from sklearn.impute import KNNImputer
KNNImputer(n_neighbors=3)
df_KNNImputer=KNNImputer(n_neighbors=3).fit_transform(df)
print(df_KNNImputer)
```

```
age fare
0 22.0 7.25
1 25.0 8.05
```

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```
[54]: #checks missing values
import pandas as pd
import seaborn as sns
from sklearn.impute import KNNImputer

#Load Titanic dataset
df=sns.load_dataset('titanic')

#forward fill for embarked
df['embarked_ffill']=df['embarked'].ffill()

#KNN Imputation for age
cols=[
    'age',
    'fare',
    'sibsp',
    'parch',
    'pclass',
    'embark_town',
    'deck',
    'embarked',
]

imputer=KNNImputer(n_neighbors=7)
df[cols]=imputer.fit_transform(df[cols])
print(df.isnull().sum())
```

```
survived    0
pclass      0
sex         0
age         0
sibsp       0
parch       0
fare        0
```

```
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embararked 2
class 0
who 0
adult_male 0
deck 688
embararked_town 2
alive 0
alone 0
embararked_ffill 0
dtype: int64

[ ]: #BACKWARD FILL
df.ffill(methods='bfill', inplace=True)
or
df.bfill(inplace=True)

[8]: import pandas as pd
import seaborn as sns
from sklearn.preprocessing import LabelEncoder

#Load titanic dataset
df=sns.load_dataset('titanic')

#Label Encoding
le=LabelEncoder()
df['sex_encoded']=le.fit_transform(
df['sex']
)

#One-hot encoding
df=pd.get_dummies(
df,
columns=['embararked']
```

```
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)

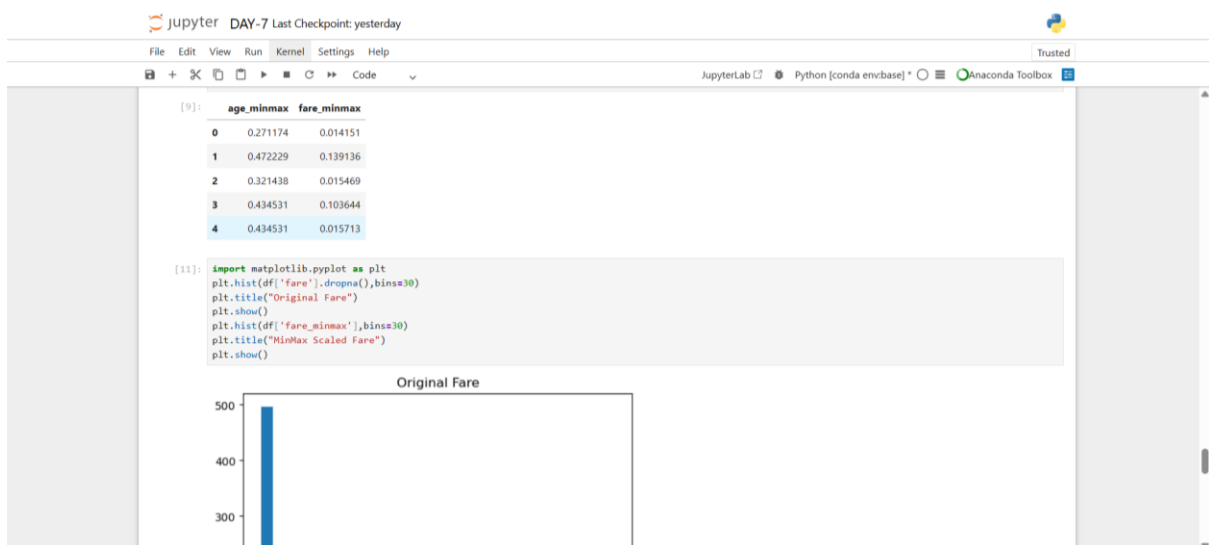
#Target Encoding
target_map=df.groupby('pclass')['survived'].mean()
df['class_target']=df['pclass'].map(target_map)
print(df.head())

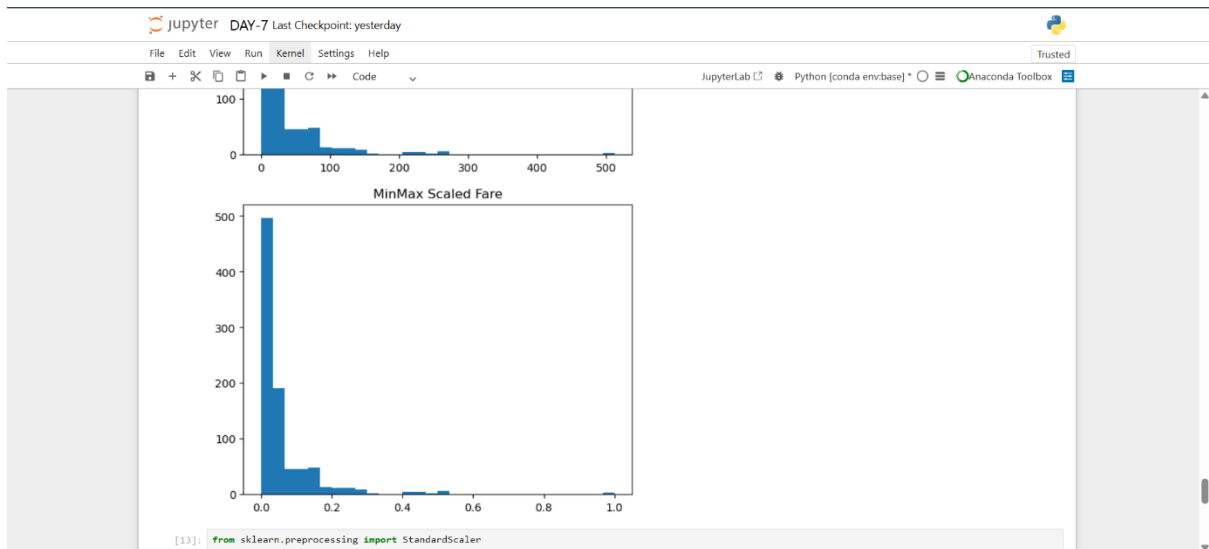
survived pclass sex age sibsp parch fare class who \
0 0 3 male 22.0 1 0 7.2500 Third man
1 1 1 female 38.0 1 0 71.2833 First woman
2 1 3 female 26.0 0 0 7.9250 Third woman
3 1 1 female 35.0 1 0 53.1000 First woman
4 0 3 male 35.0 0 0 8.0500 Third man

adult_male deck embark_town alive alone sex_encoded embarked_C \
0 True NaN Southampton no False 1 False
1 False C Cherbourg yes False 0 True
2 False NaN Southampton yes True 0 False
3 False C Southampton yes False 0 False
4 True NaN Southampton no True 1 False

embararked_Q embararked_S class_target
0 False True 0.242363
1 False False 0.629630
2 False True 0.242363
3 False True 0.629630
4 False True 0.242363

[9]: from sklearn.preprocessing import MinMaxScaler
scaler=MinMaxScaler()
df[['age_minmax','fare_minmax']]=scaler.fit_transform(
df[['age','fare']]
)
df[['age_minmax','fare_minmax']].head()
```





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```
[13]: from sklearn.preprocessing import StandardScaler
scaler=StandardScaler()
df[['age_minmax','fare_minmax']] = scaler.fit_transform(
    df[['age','fare']]
)
df[['age_minmax','fare_minmax']].head()
```

```
[13]: age_minmax fare_minmax
0    -0.530377 -0.502445
1     0.571831  0.786845
2    -0.254825 -0.488854
3     0.365167  0.420730
4     0.365167 -0.486337
```

```
[14]: from sklearn.preprocessing import RobustScaler
scaler=RobustScaler()
df[['age_robust','fare_minmax']] = scaler.fit_transform(
    df[['age','fare']]
)
df[['age_minmax','fare_minmax']].head()
```

```
[14]: age_minmax fare_minmax
0    -0.530377 -0.312011
1     0.571831  2.461242
2    -0.254825 -0.282777
```

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```
3     0.365167  0.420730
4     0.365167 -0.486337
```

```
[14]: from sklearn.preprocessing import RobustScaler
scaler=RobustScaler()
df[['age_robust','fare_minmax']] = scaler.fit_transform(
    df[['age','fare']]
)
df[['age_minmax','fare_minmax']].head()
```

```
[14]: age_minmax fare_minmax
0    -0.530377 -0.312011
1     0.571831  2.461242
2    -0.254825 -0.282777
3     0.365167  1.673732
4     0.365167 -0.277363
```

```
[ ]:
```